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BSIT 2-1

Object-Oriented Programming

MIDTERMS ASSIGNMENT #1 - Introduction to Java

1. What is Java programming language? Discuss its key features and advantage

- Java is a programming language used to create different kinds of applications such as desktop programs, web systems, and mobile apps. It was originally developed by Sun Microsystems and is now owned by Oracle Corporation. Java became very popular because it is powerful, reliable, and widely supported. It supports Object-Oriented Programming (OOP), which means it uses classes and objects to organize code properly. It is also known for being simple compared to other languages because its syntax easy to understand. Another important feature of Java is the security, it ensures reliability and security through strong memory management and exception handling. Because of these features, Java is widely used in schools and in the industry.

2. Explain the concept of platform in independence in Java. How does Java achieve platform independence?

- The meaning of Java platform-independent is that the Java compiled code (byte code) can run on all operating systems. For example, a Java program created on Windows can still run on MACOS or Linux. Java achieves this through a special process. When you write Java code, it is first compiled into something called bytecode, not directly into machine code. This bytecode is then executed by the Java Virtual Machine or JVM. The JVM acts as a translator between the program and the computer's operating system. Since different systems have their own JVM, the same bytecode can run anywhere as long as there is a JVM installed. This is why Java is known for the idea "write once, run anywhere".

3. Discuss the main components of a Java program and their roles.

- Class - this is the blueprint of the program. Every Java program must have atleast one class because all code is written inside a class.
- Main Method - This is the starting point of the program. The program begins executing from public static void main (String [ ] args).
- Methods - These are blocks of code that perform specific tasks inside the program
- Variables - These are used to store data or information that the program needs.
- Objects - these are instances of a class. They allow us to use the properties and methods defined inside the class.

4. What are the differences between JDK, JRE, and JVM in Java?

- JDK, JRE, and JVM are related to running and creating Java programs, but they have different purposes.
  - The JVM or Java Virtual Machine is the part that executes the bytecode and translates it into machine language so the computer can undertand it.
  - The JRE or Java Runtime Environment includes the JVM along with the necessary libraries and files needed to run Java Applications. This is what you need if you only want to run Java programs.
  - The JDK or Java Development Kit includes the JRE plus additional tools such as the compiler that are needed to develop and compile Java programs.
- In conclusion, the JVM runs the program, the JRE allows you to run Java applications, and the JDK is used to create and develop them.

5. Explain the difference between interpreted and compiled languages. Is Java an interpreted or compiled language?

- A compiled language is converted directly into machine code before it runs, which means the whole program is translated first and then executed. An interpred language, on the other hand, runs code line by line using interpreter without creating a seperate machine code file beforehand.

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Java is considered both compiled and interpreted. First, the Java source code is compiled into bytecode using the Java compiler. After that, the JVM interprets and executes the bytecode on the specific machine. Because of this two step process, Java is often described as a hybrid language since it uses both compilation and interpretation

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